

3(a)(i)	blood, travels through the heart twice for every one complete, circuit (of the body) / circulation or blood, travels through the heart twice and through pulmonary and systemic, circulations / systems / circuits ; A description of both circulations	1
3(a)(ii)	chordae tendineae / tendinous cords ; A tendons / 'heart strings' <i>any two from:</i> <i>idea that</i> (atrioventricular) valve, held in position / prevented from inverting, during ventricular, systole / contraction ; A AV valve <i>for atrioventricular valve</i> connect the papillary muscle to the (atrioventricular / AV), valve ; <i>ref. to</i> tension, in the chordae tendineae / AW, (during ventricular contraction) ; (allows valve to) prevent backflow of blood into atrium / ensure one way blood flow (through the heart) ; <i>ecf max 2 marks for function if structure A is a correctly named valve (atrioventricular / tricuspid / bicuspid)</i>	3

Question	Answer	Marks
3(b)(i)	<i>idea of</i> using both antibiotics makes sure all bacteria are killed ; if, resistance against one antibiotic occurs then, the other antibiotic will be effective / bacteria still susceptible to the other antibiotic or if, some bacteria are less susceptible to one antibiotic, then, the other antibiotic will be effective / bacteria still susceptible to the other antibiotic ; <i>any one from:</i> antibiotics / gentamicin and penicillin G, have different sites of action / AW ; A suggestion of how the antibiotics have different ways of killing <i>ref. to</i> mutations, qualified ; e.g. need two mutations to occur to be resistant to both antibiotics probability / AW, of these two mutations occurring in the same bacterium is very small mutations must be in different genes	2
3(b)(ii)	<i>any three from:</i> course of antibiotic is not completed ; less susceptible bacteria survive ; A not all bacteria destroyed replication / population growth, increases chance of mutation ; <i>ref. to</i> random mutation, ref to gene / allele, for resistance to gentamicin ; suggestion of outcome of different mutations ; e.g. changed ribosome structure so gentamicin cannot bind enzyme to breakdown gentamicin <i>ref. to</i> transmission of allele for resistance ; e.g. vertical transmission / described horizontal transmission / described (conjugation / transformation) e.g. frequency of allele for resistance increases in the population ; <i>when antibiotic next prescribed</i> antibiotic in the environment is the selection pressure / resistant bacteria selected for / susceptible bacteria selected against ;	3